

Chapter 4, Section 5

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1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). $\square$

2. If X is a curve of genus ≥ 2 over a field of characteristic 0, show that the group $\text{Aut } X$ of automorphisms of X is finite.

Proof. X is hyperelliptic. Let $\pi : X \rightarrow \mathbb{P}^1$ be the unique g_2^1 of X . By the Riemann-Hurwitz formula, π is ramified at $2g + 2$ points. If σ is an automorphism of X , then there exists $\tau \in \text{Aut}(\mathbb{P}^1)$ such that we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \pi \downarrow & & \downarrow \pi \\ \mathbb{P}^1 & \xrightarrow{\tau} & \mathbb{P}^1 \end{array}.$$

This means $\text{Aut}(X)$ permutes the branch points of π , so there is a group homomorphism $\text{Aut}(X) \rightarrow S_{2g+2}$. The latter is a finite group, so it remains to show it has finite kernel. If σ fixes every branch point, then τ is the identity, so σ is either the identity map, or the unique hyperelliptic involution (think in terms of function fields and the fact that the hyperelliptic involution is induced by the unique non-trivial element in $\text{Gal}(K(X)/K(\mathbb{P}^1))$).

X is nonhyperelliptic. Consider the canonical embedding $X \hookrightarrow |\mathcal{K}| \cong \mathbb{P}^{g-1}$. If σ is an automorphism of X , then there exists $\tau \in \text{PGL}(g-1)$ such that we have the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\sigma} & X \\ \downarrow & & \downarrow \\ \mathbb{P}^{g-1} & \xrightarrow{\tau} & \mathbb{P}^{g-1} \end{array}.$$

By (Ex. 4.6), X has $g(g^2 - 1)$ hyperosculation points in this embedding, and τ must permute them because linear transformations preserve incidence structures. So there is a group homomorphism $\text{Aut}(X) \rightarrow S_{g(g^2-1)}$, and since $S_{g(g^2-1)}$ is finite, it remains to show the kernel is finite. For every $g \geq 3$, $X \cap H$ can contain at most point two hyperosculation points for any hyperplane $H \subset \mathbb{P}^{g-1}$, i.e., no three hyperosculation points are collinear. It is a general fact that any automorphism in $\text{PGL}(n)$ is identity if it fixes at least $n + 2$ points in general position. Since $g(g^2 - 1) \geq g + 1$ for every $g \geq 3$, if τ fixes the hyperosculation points, it must be the identity. Hence, σ is the identity. As a corollary, any automorphism of a nonhyperelliptic is induced by an automorphism of the canonical embedding, i.e., there exists an injective map $\text{Aut}(X) \rightarrow \text{PGL}(g-1)$. $\square$

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g_3^1 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbb{P}^1$, ramified at $0, 1, \infty$, and $2g - 1 = 7$ additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of $\mathcal{M}_4$ (5.5.5).

The canonical model of a nonhyperelliptic curve of genus 4 is the complete intersection of a unique quadric surface and a cubic surface in $\mathbb{P}^3$ (5.2.2). The family of all these curves is parametrized by the space S of all complete intersections of a quadric surface and a cubic surface. We can think of S as the set of all homogenous ideals generated by a degree two and three homogenous polynomial in $k[x_1, \dots, x_4]$. So there is a morphism $S \rightarrow \mathfrak{M}_4$, whose fibres are images of the group $\mathrm{PGL}(3)$, which has dimension 15. It remains to compute the dimension of S . There is a projection map $S \rightarrow \Omega$, where Ω is the space of all irreducible quadric surfaces in $\mathbb{P}^3$. It is a locally closed subset of $H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2))$, which has dimension $\binom{3+2}{2} - 1 = 9$. The fibres of $S \rightarrow \Omega$ is the image of pullback of the linear system of cubic surfaces in $\mathbb{P}^3$, which is just $|H^0(Q, \mathcal{O}_Q(3))|$ for some quadric surface $Q \subset \mathbb{P}^2$. Note that the dimension of this space is independent of the choice of Q . We have the short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-2) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_Q \longrightarrow 0.$$

Tensoring by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(1)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow H^0(Q, \mathcal{O}_Q(3)) \longrightarrow 0.$$

By exactness, we have

$$\dim |H^0(Q, \mathcal{O}_Q(3))| = \binom{3+3}{3} - \binom{1}{3} + 1 = 15,$$

which implies the fibres of $S \rightarrow \Omega$ have dimension 15. So S has dimension $9 + 15 = 24$. Hence, $\mathfrak{M}_4$ has dimension $24 - 15 = 9$. The subspace of Ω consisting of quadric cones have dimension 8, so the space of nonhyperelliptic curves of genus 4 with only one g_3^1 is an irreducible family of dimension 8. □

5. Curves of Genus 5. Assume X is not hyperelliptic.

- (a) The curves of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection $F_2.F_2.F_2$ form a family of dimension 12.
- (b) X has a g_3^1 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.
- (c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbb{P}^2 \rightarrow \mathbb{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbb{P}^4$, with V isomorphic to $\mathbb{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g_3^1 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X . Thus V and g_3^1 are unique.

Proof.

- (a) A curve of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection of three quadrics is given by a three-dimensional subspace of $H^0(\mathbb{P}^4, \mathcal{O}_{\mathbb{P}^4}(2))$, which has dimension $\binom{4+2}{2} = 15$. Thus, the family of all these curves is parametrized by an open subset of the Grassmannian $\mathrm{Gr}(3, 15)$, which has dimension $3(15-3) = 36$. There is a natural action of $\mathrm{PGL}(4)$ on $\mathrm{Gr}(3, 15)$, and since the embedding is canonical, two three-dimensional subspaces in $\mathrm{Gr}(3, 15)$ give isomorphic curves if and only if they are in the same orbit of $\mathrm{PGL}(4)$. So there is a morphism $\mathrm{Gr}(3, 15) \dashrightarrow \mathfrak{M}_5$, whose fibres are the images of the group $\mathrm{PGL}(4)$ which has dimension $(4+1)^2 - 1 = 24$. Since any individual curve has only finitely many automorphisms (Ex. 5.2), the fibres have dimension $= 24$, and so the image of $\mathrm{Gr}(3, 15)$ has dimension $36 - 24 = 12$.
- (b) Let X be a nonhyperelliptic curve of genus 5 in its canonical embedding as a curve of degree 8 in $\mathbb{P}^4$. The converse direction is proven in (c), so it remains to show the other direction. By (5.3.3), X has a g_3^1 if and only if it has a trisecant L in this embedding. In $X \cap L = P_1, P_2, P_3$, the points are counted with multiplicity $= 8$ If P is any point in X , then

By (3.11.1), the space $V_{d,r}$ of irreducible plane curves of degree d and r nodes is irreducible and has dimension $\frac{1}{2}d(d+3) - r$. In our case, $V_{5,1}$ has dimension 19, and we have just shown $V_{5,1}$ parametrizes the space of all genus 5 curves with a g_3^1 . So there is a morphism $V_{5,1} \rightarrow \mathfrak{M}_5$, whose fibres are images of the group $\mathrm{PGL}(2)$, which has dimension 8. Hence, the family of genus 5 curves with a g_3^1 form an irreducible family of dimension $19 - 8 = 11$.

- (c) Let $\pi: V \rightarrow \mathbb{P}^2$ be the blowing up of $\mathbb{P}^2$ at the node P of $X \rightarrow X' \subset \mathbb{P}^2$ as a plane quintic curve, and let $E \subset V$ be the exceptional locus of π . Let H be any hyperplane in $\mathbb{P}^2$ so that $|2H|$ is the linear system of conics in $\mathbb{P}^2$. Using the results of (V, §3) and (V, §4), the linear system $|2H - P|$ of conics passing through P has dimension $5 - 1 = 4$ with an assigned basepoint at P . By (II, 7.17.3), the rational map $\mathbb{P}^2 \dashrightarrow |2H - P| \cong \mathbb{P}^4$ extends to a morphism $V \rightarrow \mathbb{P}^4$. It is an embedding of V into $\mathbb{P}^4$ because it corresponds to the linear system $|D|$ with $D = 2\pi^*H - E$ on V , which is very ample by (V, 4.2.2). By (V, Ex. 1.1) and (V, 3.2), the degree of V in $\mathbb{P}^4$ equals to

$$D^2 = 4H^2 + E^2 = 4 - 1 = 3.$$

The strict transform $\tilde{X}' \rightarrow X'$ is nonsingular, i.e., $\tilde{X}' \cong X$, and by (V, 3.6), it relates to X' as divisors in V in the following way

$$\pi^*X' = \tilde{X}' + 2E.$$

Using the fact that $\tilde{X} \sim 5H$ in $\mathbb{P}^2$, by (V, Ex. 1) the degree of $\tilde{X}'$ as a curve in $\mathbb{P}^4$ equals to

$$\tilde{X}' \cdot D = (\pi^*X' - 2E) \cdot (2\pi^*H - E) = 10(\pi^*H)^2 + 2E^2 = 10 - 2 = 8.$$

By the Riemann-Roch theorem, the only degree $2g - 2$ invertible sheaf with at least g sections on a curve of genus g is the canonical sheaf, so this embedding of X in $\mathbb{P}^4$ is canonical. □

- 6.** Show that a nonsingular plane curve of degree 5 has no g_3^1 . Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Proof. For the second part, it is enough to show there exists nonhyperelliptic curves of genus 6 with a g_3^1 . Consider any irreducible nonsingular curve X associated to a divisor of type $(3, 4)$ on the nonsingular quadric surface Q . It has genus $(3 - 1)(4 - 1) = 6$, and one of the families of lines on Q cuts out a g_3^1 on X . Hence, X cannot be represented as a nonsingular plane quintic curve. The canonical sheaf of Q is $\omega_Q = \mathcal{O}_Q(-2, -2)$, so by (II, 8.20), the canonical sheaf of X is

$$\omega_X \cong (\omega_Q \otimes \mathcal{O}_Q(X)) \otimes \mathcal{O}_X = \mathcal{O}_Q(1, 2) \otimes \mathcal{O}_X.$$

Since $\mathcal{O}_Q(1, 2)$ is a very ample on Q , ω_X is very ample on X . Hence, X is nonhyperelliptic (5.2). □